Table 1. Public datasets and analytical roles.

Set	BioProject	Comparison or role	Observations (n)
A01	PRJEB33360	Sepsis 67 vs ICU 79	146
A02	PRJNA691455	Sepsis 10 vs ICU 10	20
A03	PRJNA691455	Sepsis 10 vs healthy 10	20
A06	PRJNA978257	Sepsis 6 vs healthy 7	13
A05	PRJNA1010969	Sepsis 18 vs trauma 18; primary external	36
A04	PRJNA1010969	Sepsis 18 vs healthy 17; supportive external	35
S01/S02	PRJEB33360	Mixed models, 43 patients each	130 / 127
S03	PRJNA691455	Longitudinal, 10 patients	29
S04	PRJNA430161	Intervention support, 9 patients	26
S05	PRJNA797231	Myocardial dysfunction support	34
S06	PRJNA912621	Cholestasis, 20 patients	58

A02/A03 and A04/A05 share sepsis participants. S01/S02 totals include baseline records. Observation counts must not be summed as unique patients. ICU, intensive care unit.

Table 2. Community-level differences in discovery analysis sets.

Comparison	PERMANOVA R^2	BH FDR
A01	0.0147	0.044
A02	0.0537	0.401
A03	0.1449	0.007
A06	0.1311	0.044

PERMANOVA R^2 is reported as a proportion. FDR, Benjamini-Hochberg false discovery rate. Full alpha-diversity, dispersion and genus meta-analysis results are provided in Supplementary Data 1.

Table 3. SDI development and external evaluation.

Evaluation	AUC	95% bootstrap CI
Pooled development, apparent	0.731	0.656–0.806
A01 development	0.733	0.645–0.812
A02 development	0.690	0.420–0.900
A05 primary external	0.272	0.105–0.451
A04 supportive external	0.201	0.046–0.373

Equal-weight SDI with the development direction retained. The fixed development threshold is −1.32849067. A05 and A04 are the primary and supportive external comparisons from one BioProject. CI, confidence interval.

Table 4. Longitudinal Shannon diversity tests.

Analysis	Omnibus term	P value	BH FDR
T01, 130 observations / 43 patients	Time	0.00237	0.0126
T02, 127 observations / 43 patients	Time	0.00446	0.0214
T03, 29 observations / 10 patients	Time	0.0578	0.318
T06, 58 observations / 20 patients	Time × cholestasis	0.00931	0.266

FDR families include all non-intercept omnibus terms across metrics within each analysis ID. The T01 day-7 coefficient was −0.772 (95% CI −1.144 to −0.401); this contrast is distinct from the omnibus test. Further supportive results are provided in Supplementary Data 1.

Table 5. Project-specific DADA2 filtering parameters.

BioProject	Read mode	truncLen / trimLeft	maxEE
PRJEB33360	Single end	450 / 0	2
PRJNA691455	R1 only	250 / 17	2
PRJNA978257	Paired	0,0 / 0,0	2,5
PRJNA1010969	Paired	220,160 / 19,20	2,3
PRJNA430161	Paired	0,0 / 0,0	2,15
PRJNA797231	Paired	240,240 / 0,0	2,2
PRJNA912621	Paired	0,0 / 0,0	2,5

All projects use maxN = 0, truncQ = 2 and rm.phix = TRUE. Paired values refer to forward and reverse reads. A truncLen of zero indicates no fixed-length truncation. Complete processing arguments are available in the archived scripts.